

Adapting to Drift: Weather-Pattern Experts for Short-Term Photovoltaic Power Forecasting

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ABSTRACT

Photovoltaic (PV) power forecasting is often developed under an implicit assumption of stationarity, yet the weather–power relationship evolves over time and induces distribution shifts commonly known as concept drift. Existing PV power forecasting models either ignore this issue entirely or address it in a limited and insufficient manner. To better understand this non-stationarity, we distinguish between internal and external drift, which motivate the key components of our design. Therefore, we present ADrift, a drift-aware forecasting framework that integrates patch-based temporal modeling, weather-guided representation learning, and lightweight online adaptation. To model internal drift, the backbone employs prototype-guided weather experts that capture diverse meteorological patterns within each input window. To cope with external drift, a learnable adapter updates its parameters through a temporal gap attention mechanism that enables targeted adjustments to the model. In addition, a proactive update strategy further mitigates supervision delays under rapidly changing conditions. Experiments on three real-world PV datasets show that ADrift consistently improves forecasting accuracy over static and online-learning baselines, demonstrating its potential for practical deployment under evolving weather conditions.

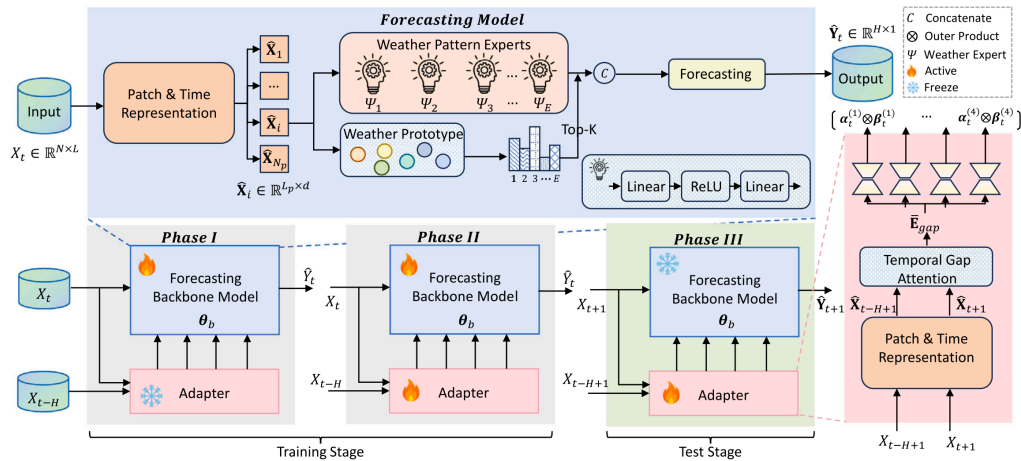


Figure 1: Model Structure of ADrift.